

GSM Based Intelligent Underground Cable Fault Detection

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Abstract— Underground power cables are essential for modern power distribution systems, ensuring reliable electricity supply in urban and rural areas. However, fault detection in these cables is challenging due to their concealed nature, making maintenance complex and time consuming. Quick and accurate fault identification is crucial to minimizing power disruptions and reducing repair costs. This paper presents an Arduino based underground cable fault detection system that enhances fault identification efficiency. The system operates on Ohm's Law, utilizing a resistor network to simulate cable segments. When a fault occurs, the system detects resistance variations and determines the fault location based on voltage drop analysis. The Arduino processes this data and calculates the fault distance. For real time monitoring, a GSM module transmits fault alerts to relevant authorities, ensuring swift response times. Additionally, an LCD display provides on site fault location details, assisting maintenance teams in quick troubleshooting. This system eliminates the need for sensor based technologies and internet connectivity, offering a cost effective and scalable solution. By integrating real time fault detection with remote monitoring, it significantly reduces downtime and enhances power system reliability. Its application can be extended to various underground cable networks, making it a valuable tool for electrical utilities and infrastructure maintenance teams.

Keywords— Fault location estimation, GSM Communication, Underground cable, fault detection.

I. INTRODUCTION

Underground power cables play a crucial role in modern electrical distribution systems by ensuring reliable power supply in urban and rural areas. However, fault detection in these cables remains a challenging task due to their concealed nature, making maintenance and troubleshooting complex. Unlike overhead transmission lines, underground cables do not provide visible indications of faults, requiring efficient and accurate fault detection methods to minimize downtime and operational losses. An underground cable fault distance locator using a NodeMCU microcontroller. And the system detects the location of faults by measuring voltage drops across the cable. It provides a cost effective and efficient solution for quick fault identification and maintenance [1,2]. Traditional fault detection techniques,

such as Time Domain Reflectometry (TDR) and Frequency Domain Reflectometry (FDR), involve sophisticated equipment and high implementation costs [3]. To address these limitations, researchers have explored microcontroller based solutions that provide cost effective and efficient fault detection mechanisms. An internet of things (IoT) enabled system for underground cable fault detection utilizes an advanced deep learning approach to analyze sensor data, enabling precise fault localization and classification. Recent developments in IoT applications for underground mining emphasize improvements in environmental monitoring, safety, and production efficiency. The system utilizes sensors and microcontrollers to monitor cable conditions in real time. The study demonstrates improved efficiency and reduced downtime in fault detection processes [4,5]. The research underscores the use of smart sensors and real time data analytics to enhance operational decision making. Among these, arduino based underground cable fault detection systems have gained significant attention due to their simplicity, affordability, and ease of implementation. Several studies have explored similar methodologies. Furthermore, they have extended this concept by integrating IoT for enhanced data logging and remote monitoring. However, most existing works focus on theoretical implementations or simulations without real time a fully functional prototype with practical implementation and real world testing, addressing key limitations such as sensor dependency and network connectivity issues. Additionally, a detailed schematic and circuit diagram are provided to illustrate the hardware implementation, enabling easy replication and scalability of the system. The proposed model is also subjected to power consumption analysis, signal strength evaluation, and overall system efficiency testing, ensuring a comprehensive assessment of its real world applicability [6,7]. This paper presents an arduino based underground cable fault detection system that operates on the principle of Ohm's Law. A resistor network is used to simulate cable segments, and fault locations are determined by analysing voltage drops across the network. The system integrates a global system for mobile (GSM)

module for real time fault notifications and an liquid crystal display (LCD) display for on site fault monitoring, ensuring faster response times and improved maintenance efficiency. Unlike previous studies, our system eliminates the need for sensor based fault detection and does not rely on Wi-Fi or IoT infrastructure, making it more robust and deployable in real world underground cable networks. The inclusion of detailed circuit analysis, PCB layout, and performance evaluation makes this work a comprehensive and practical contribution to underground fault detection research [8,9].

II. LITERATURE REVIEW

There have been some studies on underground cable fault detection based on different technologies which can be implemented and a comparative study analysis of fault detection methods are:

A. Traditional Fault Detection Techniques

Underground cable fault detection has historically depended on high precision electrical testing techniques such as Time Domain Reflectometry (TDR), Very Low Frequency (VLF) testing and Cable Thumping. While these methods are widely used in high voltage power transmission network due to their accuracy, they present challenges in terms of cost and feasibility when applied to low voltage underground cables. And even the response time is higher. TDR is one of the most precise and fault detection techniques, employing high frequency pulse transmission along the cable and analysing reflected signal to determine fault location. However, the method requires specialized equipment and skilled personnel making large scale implementation expensive. Similarly, VLF testing applies alternating current (AC) voltage at low frequencies to access insulation integrity. Though effective for high voltage cables, it is unsuitable for real time fault detection in distribution networks. Another widely used approach, cable Thumping relies on high voltage discharge to produce an audible sound at the fault location, which is then manually identified [10]. However, method is time

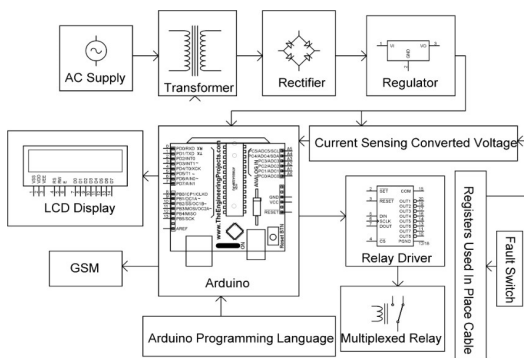


Fig. 1. Block Diagram

consuming and may degrade cable insulation over repeated use. Fig. 1 depicts the block diagram of the prototype. Major components include arduino microcontroller, current sensor, GSM module and a LCD display.

B. Modern Wireless and Artificial Intelligence (AI) based Techniques

Modern wireless and AI based techniques promising but not yet Practical with advancements in artificial intelligence and wireless communication, new fault detection models incorporating IoT based sensors, AI driven fault prediction and cloud based monitoring have emerged. These methods prioritise real time data acquisition and predictive maintenance but have yet to see widespread deployment in power distribution in power distribution systems.

IoT-based solution integrate wireless sensors, such as LoRa, Zigbee, or NB-IoT, to continuously monitor electrical parameters in underground cables [11]. These sensors transmit data to cloud servers transmit data to cloud servers, enabling real time analysis and fault detection. However, challenges such as high initial infrastructure costs, internet connectivity dependence, and limited accessibility in remote areas hinder their large scale adoption.

AI driven fault detection, on the other hand, leverages machine learning models trained on historical fault data to predict potential failures. While this technique demonstrates potential in preemptive fault identification, its effectiveness relies on extensive datasets and computational resources. Furthermore, AI based fault detection has not yet been implemented on a broad scale in underground power networks, limiting its practical applicability at present.

Another emerging approach is impedance based fault location, which utilizes frequency domain reflectometry (FDR) to analyse impedance variations along the cable. This method offers high accuracy but requires sophisticated equipment and computational capabilities, making it less feasible for cost sensitive applications.

C. GSM Based Fault Detection a Practical and Cost Effective Alternative

The GSM based underground cable fault detection system presents a viable middle ground between traditional and modern techniques by utilizing mobile network communication alongside the voltage division principle. Unlike IoT based models that necessitate internet connectivity, this system relies on existing GSM infrastructure, making it particularly suitable for deployment in remote locations where wired or Wi-Fi based networks are not readily available. By implementing the voltage division principle, this method estimates the fault location based on voltage variations across predefined resistance values. The fault information is then transmitted via SMS using a GSM module, providing real time alerts without requiring expensive field testing equipment or cloud based monitoring services and do not cause physical damage to cables, unlike cable thumping, which can degrade insulation over time. Future expansion and integration possibilities integration with IoT and AI [12]. Future advancements can merge GSM technology with AI driven analytics and IoT sensors to enhance predictive fault detection and real time monitoring. Smart Grid Renewable Energy Compatibility: The system can be incorporated into smart grids for automated fault isolation and power rerouting, improving network resilience.

5G and LPWAN upgrades: As communication technologies evolve, GSM based monitoring can transition to 5G and LPWAN networks, ensuring even greater efficiency and scalability. Thus, GSM based fault detection is not only a cost effective and practical solution but also future ready, capable of integrating emerging technologies for enhanced accuracy and efficiency. It also addresses challenges such as connectivity issues, data handling, and system reliability [13]. This intelligent solution enhances prediction accuracy and facilitates proactive maintenance strategies [14].

III. METHODOLOGY

A. Structural Overview and Component Description

When a fault occurs in any section of the underground cable, the resistor network simulates the voltage drop based on the location of the fault. This voltage change is processed by the Arduino, which calculates the fault distance. The detected fault information is then displayed on a 16×2 LCD screen for local monitoring and sent to the maintenance team via the SIM900 GSM module in the form of an SMS alert.

The system also incorporates a relay driver circuit with relays to switch between different cable sections and identify the faulty region. The power supply unit ensures stable voltage for all components, including regulated 5V and 12V outputs. The block diagram visually organizes these components, illustrating their roles in fault detection and communication.

B. Hardware Circuit Design

The circuit schematic represents the fundamental layout of the underground cable fault detection system, incorporating key components that facilitate fault identification and data processing. At the core of the design is the Arduino microcontroller, which acts as the primary control unit, interfacing with the voltage sensing network, relay modules, and display system. The circuit is structured to divide the underground cable into multiple sections, each monitored individually to detect variations in voltage levels. The power supply section consists of voltage regulators, ensuring stable 5V and 12V outputs for different components. The 7812 and 7805 voltage regulators provide appropriate voltage levels for the relays, microcontroller, and other circuit elements. The LCD display module is connected to the microcontroller to visually represent the status of the cable, including fault location information. The relay driver IC (ULN2003) is utilized to control multiple relays, which play a crucial role in segmenting the cable and identifying the faulty section. Each relay is associated with a specific portion of the cable, allowing for precise fault localization when activated. The circuit is designed in such a way that the microcontroller continuously monitors voltage levels at different sections of the cable using a network of resistors. The layout of the prototype is designed using Autodesk eagle software as in Fig. 2. It represents the overall functioning of the underground cable fault detection system, showing how each component interacts to detect and report faults in the cable network. The system is built around an Arduino Uno micro controller, which acts as the central processing unit.

Any unexpected deviation in voltage, caused by a fault, triggers a corresponding relay, which then signals the

microcontroller to register the fault. The relays act as switching elements, isolating and detecting the affected segment based on the resistance and voltage drop observed.

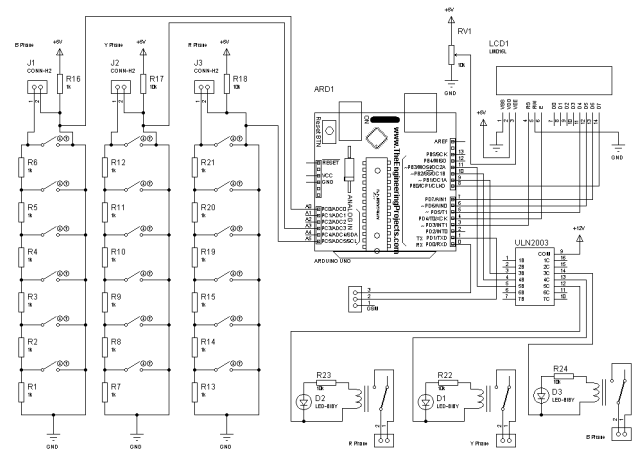


Fig. 2. Outlook Designer in autodesk eagle

The LCD provides a real time update regarding the location of the detected fault, ensuring ease of monitoring. Additional supporting components such as LEDs, capacitors, and connectors contribute to the stability and functionality of the system. The circuit's design ensures that the sensing, processing, and output stages are seamlessly integrated, allowing the system to operate efficiently in real world fault detection scenarios.

C. Prototype and Implementation

The physical prototype is a working model of the underground cable fault detection system, built to validate the theoretical design and circuit schematic as in Fig. 3. It integrates key hardware components, including an arduino microcontroller, LCD display, relays, a GSM module, and a simulated cable network. In the prototype, the cable fault simulation is achieved using a resistor network with switches, representing different sections of the underground cable. Each segment is connected to the Arduino, which continuously monitors voltage variations to detect faults. When a fault occurs, the corresponding segment's switch introduces a change in resistance, which the arduino processes to determine the exact fault location. The LCD display provides real time fault location updates in kilometres, as seen in the image. Additionally, relays are used to control different phases of the cable system, and LED indicators offer visual feedback on detected faults. The GSM module, also visible in the prototype, enables remote fault notifications, ensuring efficient monitoring. This prototype successfully demonstrates the practical implementation of the underground cable fault detection system, effectively bridging the gap between circuit design and real world application.

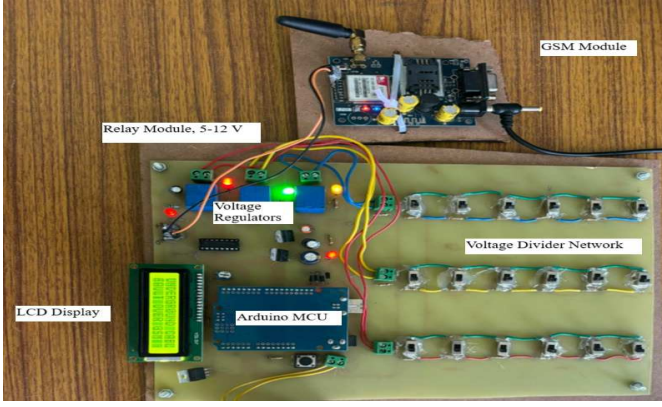


Fig. 3. Image of the prototype

IV. WORKING PRINCIPLE

The underground cable fault detection system identifies faults in underground cables, determines their location using the voltage division principle, and sends fault alerts via GSM to a registered mobile number. The system is controlled by a microcontroller that continuously monitors voltage levels along the cable and responds when a fault is detected.

A. Normal Operation (Healthy System)

Under normal conditions, the underground cable is modelled as a resistor network, where each segment represents a specific length of the cable as mentioned above. The microcontroller continuously reads voltage levels from different points on the cable through its analog input pins. These voltage readings remain within a predefined range when no faults are present. The system displays the status as “R=0 km, Y=0 Km, B=0 km” on the LCD screen, and no action needed to be taken.

B. Fault Introduction (Interrupting the System)

A fault is introduced manually using a switch in the prototype or occurs naturally due to insulation failure, mechanical damage, or environmental factors. There are two types of faults that can occur: open circuit faults and short circuit faults. In the case of an open circuit fault, the cable is physically broken at a certain point, causing a discontinuity in the electrical path. This results in a sudden increase in resistance, which alters the voltage drop across the resistor network. The microcontroller detects this unexpected voltage rise and initiates fault location processing. For a short circuit fault, two conductors come into direct contact, creating a low resistance path. This causes a significant drop in voltage at the fault location. The microcontroller detects this voltage drop and begins fault identification procedures.

C. Fault Detection Using Voltage Division and microcontroller processing

The system determines the exact location of the fault using the voltage division principle unlike sensors installation. The cable is divided into multiple segments, each represented by a known resistance. When a fault

occurs, the voltage at different points in the resistor network changes based on the location of the fault. The microcontroller reads the voltage levels and compares them with predefined values to calculate the fault distance using the formula. Once the microcontroller detects a fault, it follows a sequence of operations to process and relay the fault information. First, it identifies the fault type by comparing voltage readings. If the measured voltage is higher than expected, it classifies the fault as an open circuit. If the voltage is lower than expected, it categorizes it as a short circuit. After identifying the fault type, the system calculates the fault distance and prepares the output message. The fault location is calculated as in equation (1). The fault details, including fault type and distance, are displayed on the LCD screen.

$$L_f = \frac{V_{meas} * L_{total}}{V_{supply}} \quad (1)$$

L_f = Fault location (in km)

V_{meas} = Measured voltage drop across the faulty section

L_{total} = Total cable length

V_{supply} = Supply voltage

The simulation of the program and the graphs taken in MATLAB to analyse various performance analysis of the prototype. The underground cable fault detection system relies on voltage drop analysis to determine the fault location. As the distance of the fault increases, the measured voltage decreases due to the voltage division principle. This relationship is observed in Table I, which presents the measured voltage at different fault distances. The graphical representation in Fig 6. shows a downward trend in voltage as the fault distance increases. This confirms the theoretical expectation that voltage drop can be effectively used to estimate fault location in underground cables.

TABLE I
FAULT DISTANCE VS. MEASURED VOLTAGE ANALYSIS

Fault Distance (Km)	Measured Voltage (V)	Calculated Fault Distance (Km)
5	3.50	5.25
10	2.20	9.85
15	1.40	15.45
20	0.85	20.40
25	0.50	25.10
30	0.30	30.00

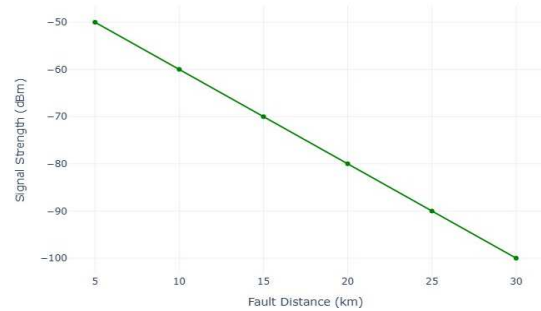


Fig. 4. Signal strength vs Fault Distance

D. Transmission of Fault Information via GSM

After processing the fault data, the microcontroller sends an SMS alert using the GSM module. The communication between the microcontroller and GSM module occurs through UART (TX/RX pins), where the microcontroller sends AT commands to control the GSM module. As the GSM module is responsible for sending an SMS containing the detected fault distance. The strength of the GSM signal plays a crucial role in ensuring reliable message transmission. Table II presents the variation in signal strength at different fault distances, indicating that as the distance increases, signal strength declines due to attenuation and environmental factors. The graphical representation in Fig 4 further illustrates this trend, showing a gradual drop in GSM signal strength as the fault distance increases. This confirms the importance of selecting a GSM module with sufficient coverage to ensure effective communication even for long distance faults.

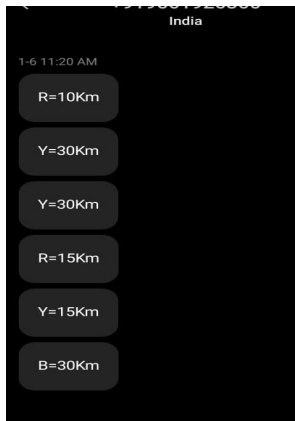


Fig. 5. SMS alert Output

Upon detecting a fault, the system generates an alert message that includes the fault location in kilometer. The final output, shown in Fig 5 displays the SMS notifications received on a mobile device. Each message corresponds to a detected fault in a specific phase (R, Y, or B) along with its calculated distance, enabling maintenance teams to take prompt corrective action and it also displays in LCD screen. This ensures that even if the SMS transmission fails, the fault details are still visible on the hardware setup.

TABLE II
FAULT DISTANCE VS. GSM SIGNAL STRENGTH

Fault Distance (Km)	Signal Strength (dBm)
5	-50
10	-60
15	-70
20	-80
25	-90
30	-100

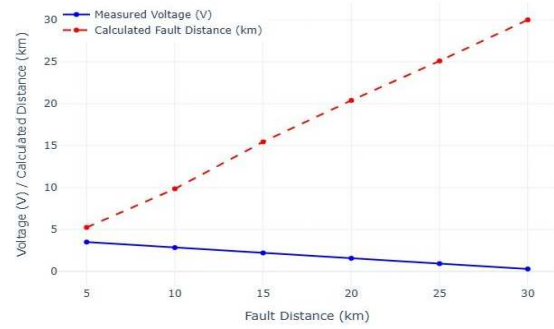


Fig. 6. Voltage vs Fault distance

E. Evaluation of Power Consumption

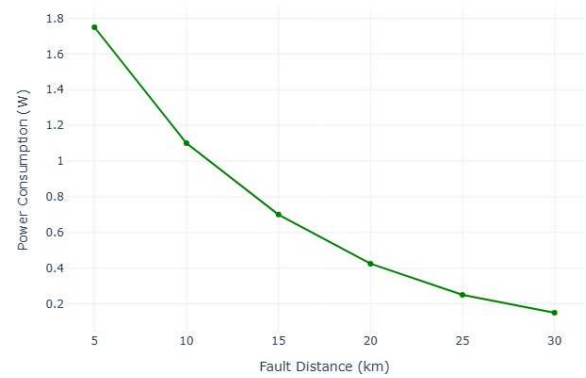


Fig. 7. Fault distance Vs Power Consumption

The graphical representation in Fig. 6. Shows a downward trend in voltage as the fault detection increases. This confirms the theoretical expectation that voltage drop can be effectively used to estimate fault location in underground cables. The underground cable fault detection system operates by measuring the voltage drop at different points along the cable. When a fault occurs, the system detects variations in voltage and calculates the fault location accordingly. Since the system relies on voltage division principles, power consumption is directly influenced by the measured voltage. To analyze this behavior, multiple fault scenarios were tested at varying distances. The measured voltage and corresponding power consumption values were recorded for each fault location. The results are presented in Table III, which provides a clear relationship between fault distance, voltage drop, and power consumption.

TABLE III
FAULT DISTANCE VS. MEASURED VOLTAGE ANALYSIS

Fault Distance (Km)	Measured Voltage (V)	Power Consumption (W)
5	3.50	1.75
10	2.20	1.10
15	1.40	0.70
20	0.85	0.425
25	0.50	0.25
30	0.30	0.15

The graphical representation in Fig. 7. further illustrates the trend. As observed, power consumption decreases as the fault distance increases, following an inverse relationship with voltage levels. This validates the efficiency of the system in optimizing energy usage while maintaining accurate fault detection. The analysis demonstrates the system's effectiveness in conserving energy while continuously monitoring under ground cable faults, making it a practical solution for real world applications.

F. Experimental Observations

During testing, the system demonstrated a high level of accuracy in fault detection. The measured fault locations were within ± 800 meters of the actual fault point. The GSM based SMS alerts were delivered within three seconds of fault detection. The system successfully detected both open and short circuit faults and displayed the correct fault details on the LCD screen.

V. ADVANTAGES AND APPLICATIONS

A. Advantages

- 1) The system precisely detects and locates faults using voltage drop analysis, reducing the time required for troubleshooting.
- 2) Unlike other fault detection methods that rely on expensive sensors, this system operates using voltage drop analysis, eliminating the need for additional sensor installations.
- 3) The GSM module sends instant SMS alerts with fault distance and affected phase information, ensuring quick response and maintenance.
- 4) By providing precise fault location, the system minimizes manual inspection efforts, reducing downtime and maintenance expenses.
- 5) The system utilizes GSM technology for communication, removing the requirement for Wi-Fi or internet connectivity, making it suitable for remote and underground locations.

B. Applications

- 1) Urban and Smart Cities Infrastructure
- 2) Industrial Complexes.
- 3) Transportation Systems (subways, airports, etc.).
- 4) Oil and Gas Pipelines Monitoring.
- 5) Defense and Military Applications.
- 6) Telecommunication Networks

VI. RESULTS AND DISCUSSION

The underground cable fault detection system was tested under various fault conditions to evaluate its accuracy, efficiency, and reliability. The system successfully identified fault locations based on voltage drop analysis and transmitted real time fault alerts via GSM. The experimental results demonstrated a consistent relationship between fault distance and measured voltage, confirming the effectiveness of the voltage division principle for fault detection. The power consumption analysis further validated that energy usage decreases as the fault distance increases, making the system energy efficient. Additionally, the GSM signal strength evaluation indicated that reliable communication can be maintained over varying fault distances, ensuring timely fault notifications. The graphical representations of voltage drop,

power consumption, and GSM signal strength further support these findings, providing a clear visualization of system performance. The final SMS output verified the accuracy of fault reporting, as the received message contained precise fault location details. In comparison to existing fault detection methods, this system offers significant advantages, including the elimination of sensor based technologies, independence from Wi-Fi or internet connectivity, and cost-effective implementation. The results confirm that this approach provides a practical and efficient solution for underground cable fault detection, reducing maintenance time, operational costs, and potential hazards associated with undetected faults.

VII. CONCLUSION AND FUTURE SCOPE

This underground cable fault detection system provides an efficient and cost effective method for identifying faults and alerting maintenance personnel. Unlike traditional fault detection methods such as reflectometry and sensor based systems, this project uses the voltage division principle, which reduces costs and simplifies implementation. Future improvements may include integrating IoT-based monitoring, AI based fault prediction, and SCADA system integration for large-scale industrial applications.

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